

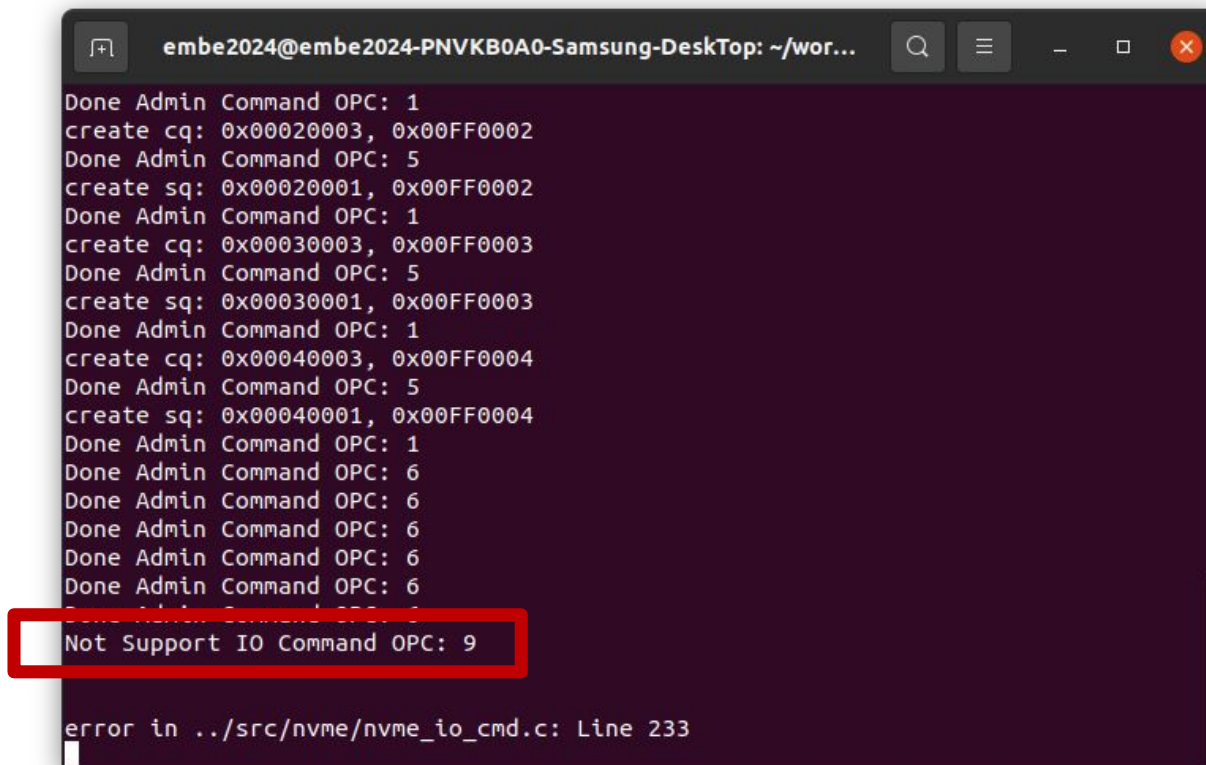


Project #4 Task1 Guide

Embedded System Software (CSE4116-01)

Task1 - 문제점

- `sudo mkfs.ext4 -F /dev/nvme0n1`
 - `mkfs`: 논리적 파티션 위에 파일시스템 (`ext4` 등)을 생성하는 명령어
 - 블록 초기화 등을 위해 NVMe DSM 명령으로 `discard`(TRIM) 요청을 전송



A terminal window titled "embe2024@embe2024-PNVKB0A0-Samsung-DeskTop: ~/wor..." displays a series of NVMe DSM commands and their responses. The commands include "create cq" and "create sq" for various IDs. The responses are "Done Admin Command OPC: 1" for the first three and "Done Admin Command OPC: 5" for the next three. The last three responses are "Done Admin Command OPC: 6". The final response, "Not Support IO Command OPC: 9", is highlighted with a red rectangle. Below this, the error message "error in ../src/nvme/nvme_io_cmd.c: Line 233" is visible.

```
Done Admin Command OPC: 1
create cq: 0x00020003, 0x00FF0002
Done Admin Command OPC: 5
create sq: 0x00020001, 0x00FF0002
Done Admin Command OPC: 1
create cq: 0x00030003, 0x00FF0003
Done Admin Command OPC: 5
create sq: 0x00030001, 0x00FF0003
Done Admin Command OPC: 1
create cq: 0x00040003, 0x00FF0004
Done Admin Command OPC: 5
create sq: 0x00040001, 0x00FF0004
Done Admin Command OPC: 1
Done Admin Command OPC: 6
Done Admin Command OPC: 6
Done Admin Command OPC: 6
Done Admin Command OPC: 6
Done Admin Command OPC: 6
Done Admin Command OPC: 6
Not Support IO Command OPC: 9
error in ../src/nvme/nvme_io_cmd.c: Line 233
```

Task1 - 해결책

1. Cosmos+ OpenSSD 펌웨어 수정

- a. discard/TRIM 요청에 대한 올바른 처리를 할 수 있게끔 **펌웨어**에 해당 명령을 구현함.

2. mkfs 옵션을 통한 우회

- a. `sudo mkfs.ext4 -F -K /dev/nvme0n1`
- b. `sudo mkfs.ext4 -F -E nodiscard /dev/nvme0n1`

3. **NVMe Driver 레벨**에서 특정 장치에 대한 discard(TRIM) 기능 비활성화

- a. **device quirk**: 스펙을 어기거나 버그가 있는 특정 하드웨어를 위한 예외 처리 플래그

Task1 - device quirk

- linux-5.15.190/drivers/nvme/host/nvme.h
 - 특정 장치를 위한 quirks 비트마스크를 추가함

```
54  enum nvme_quirks {
55      /*
56       * Prefers I/O aligned to a stripe size specified in a vendor
57       * specific Identify field.
58       */
59      NVME_QUIRK_STRIPE_SIZE          = (1 << 0),
60
61      /*
62       * The controller doesn't handle Identify value others than 0 or 1
63       * correctly.
64       */
65      NVME_QUIRK_IDENTIFY_CNS        = (1 << 1),
66
```

```
167
168  // embe-prj4
169  NVME_QUIRK_NO_DISCARD              = (1 << 21),
170  };
```

Task1 - device quirk

- linux-5.15.190/drivers/nvme/host/core.c
 - Cosmos+ OpenSSD를 식별하는 헬퍼 추가

```
23 // embe-prj4
24 #include <linux/string.h>
```

```
85 // embe-prj4
86 static bool nvme_is_cosmos_openssd(const struct nvme_id_ctrl *id)
87 {
88     return !strncasecmp(id->mn, "COSMOS+", 7);
89 }
90
```

Task1 - device quirk

- linux-5.15.190/drivers/nvme/host/core.c
 - Cosmos+ OpenSSD 보드일 때, quirk 비트 켜기

```
2972 static int nvme_init_identify(struct nvme_ctrl *ctrl)
2973 {
2974     struct nvme_id_ctrl *id;
2975     u32 max_hw_sectors;
2976     bool prev_apst_enabled;
2977     int ret;
2978
2979     ret = nvme_identify_ctrl(ctrl, &id);
2980     if (ret) {
2981         dev_err(ctrl->device, "Identify Controller failed (%d)\n", ret);
2982         return -EIO;
2983     }
2984
2985     if (id->lpa & NVME_CTRL_LPA_CMD_EFFECTS_LOG) {
2986         ret = nvme_get_effects_log(ctrl, NVME_CSI_NVM, &ctrl->effects);
2987         if (ret < 0)
2988             goto out_free;
2989     }
2990
2991     if (!(ctrl->ops->flags & NVME_F_FABRICS))
2992         ctrl->cntlid = le16_to_cpu(id->cntlid);
2993
2994     if (!ctrl->identified) {
2995         unsigned int i;
2996
2997         // embe-prj4
2998         if (nvme_is_cosmos_openssd(id))
2999             ctrl->quirks |= NVME_QUIRK_NO_DISCARD;
3000    }
```

Task1 - device quirk

- linux-5.15.190/drivers/nvme/host/core.c
 - 분기를 만들어서 discard 명령어 비활성화 하기

```
2916 static int nvme_init_non_mdts_limits(struct nvme_ctrl *ctrl)
2917 {
2918     struct nvme_command c = { };
2919     struct nvme_id_ctrl_nvm *id;
2920     int ret;
2921
2922     // embe-prj4
2923     if (ctrl->oncs & NVME_CTRL_ONCS_DSM &&
2924         !(ctrl->quirks & NVME_QUIRK_NO_DISCARD)) {
2925         ctrl->max_discard_sectors = UINT_MAX;
2926         ctrl->max_discard_segments = NVME_DSM_MAX_RANGES;
2927     } else {
2928         ctrl->max_discard_sectors = 0;
2929         ctrl->max_discard_segments = 0;
2930     }
2931 }
```

Task1 - device quirk

- linux-5.15.190/drivers/nvme/host/core.c
 - discard와 관련된 다음 작업 가기 전 조기 return

```
2944 // embe-prj4
2945 if (nvme_ctrl_limited_cns(ctrl) || (ctrl->quirks & NVME_QUIRK_NO_DISCARD))
2946     return 0;
2947
2948 id = kzalloc(sizeof(*id), GFP_KERNEL);
2949 if (!id)
2950     return -ENOMEM;
```